

Testing For Relationships

CMPSC 105 – Data Exploration



ALLEGHENY COLLEGE

Goals

- Analyses for one continuous variable
- Analyses for two continuous variables

Analyses for one continuous variable

Find sample mean

- consider a continuous variable
- finding the mean and std are the most basic forms of analysis

Hypothesis test if the sample mean is not likely 0

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 y = np.array([1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 6, 7.1, 9.9, 10])
5
6 y_bar = np.mean(y) # sample mean
7 SEM = np.std(y, ddof=1) / (len(y) ** 0.5) # standard error of the mean
8 score = ( y_bar - 0 ) / SEM
9 z_crit = 1.96 # for 95% boundary with two tails, recall z = 1.96
10
11 print(score)
12 print('probability of this score is...?')
```

5.957419483045198

probability of this score is...?

Alternative using code

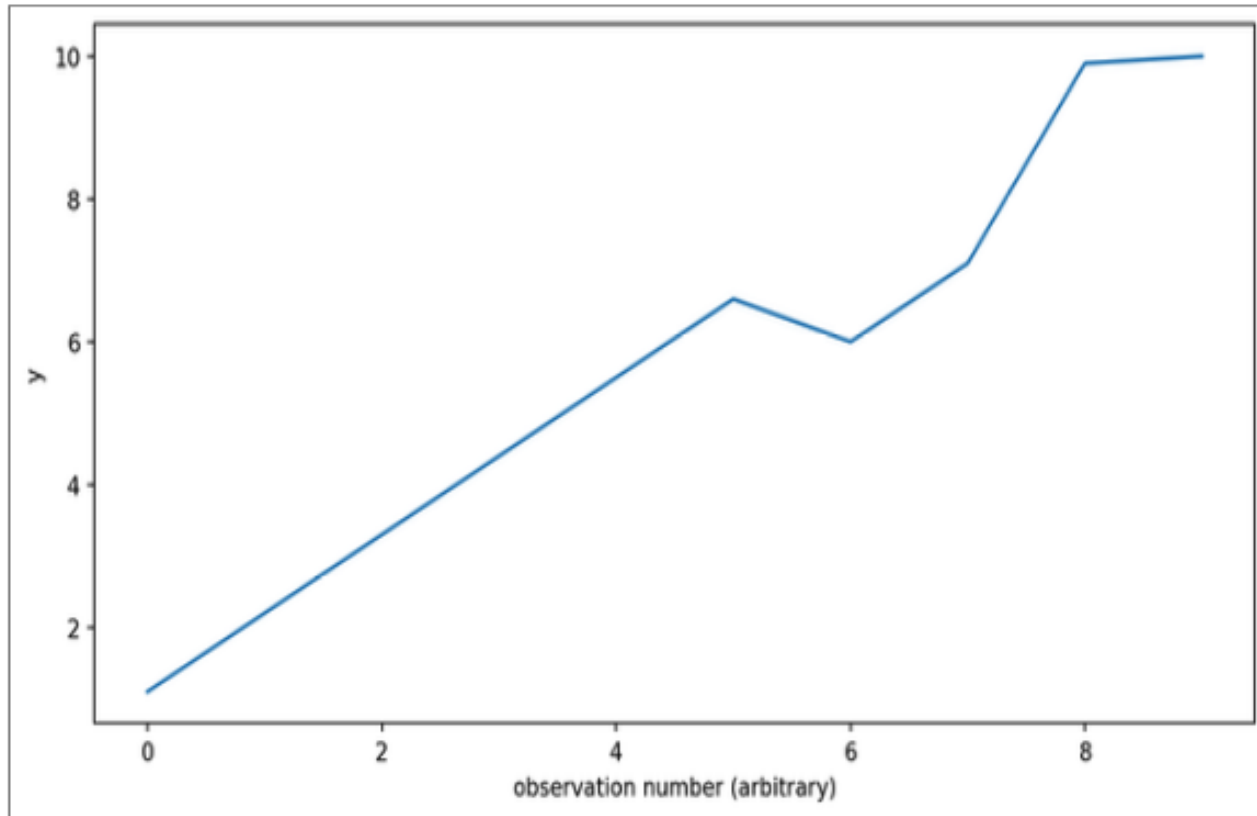
```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 import scipy
4
5 y = np.array([1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 6, 7.1, 9.9, 10])
6
7 score, probability = scipy.stats.ttest_1samp(y, 0)
8
9 print(score)
10 print(probability)
```

Examine the data

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 category = np.array(['control', 'control', 'control', 'control', 'control', 'test', 'test', 'test', 'test']
5 y = np.array([1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 6, 7.1, 9.9, 10])
```

```
1 plt.figure()
2 plt.plot(y)
3 plt.ylabel('y')
4 plt.xlabel('observation number (arbitrary)')
5 plt.show()
```

Examine the data

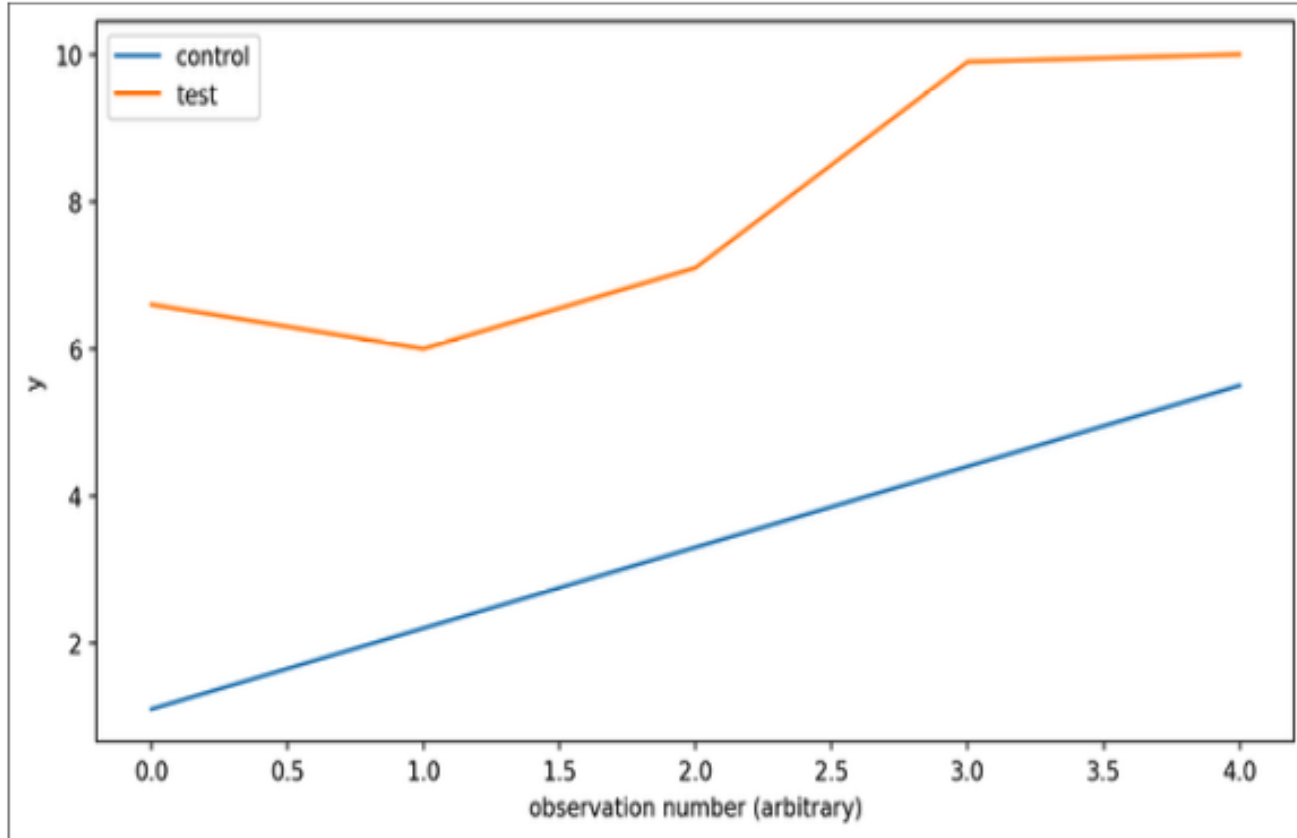


Breaking up a variable into categories

- what if I split y into control and test
- then the mean in each category could be compared

```
1 plt.figure()
2 plt.plot(y[category=='control'])
3 plt.plot(y[category=='test'])
4 plt.legend(['test', 'control'])
5 plt.ylabel('y')
6 plt.xlabel('observation number (arbitrary)')
7 plt.show()
```

Breaking up a variable into categories

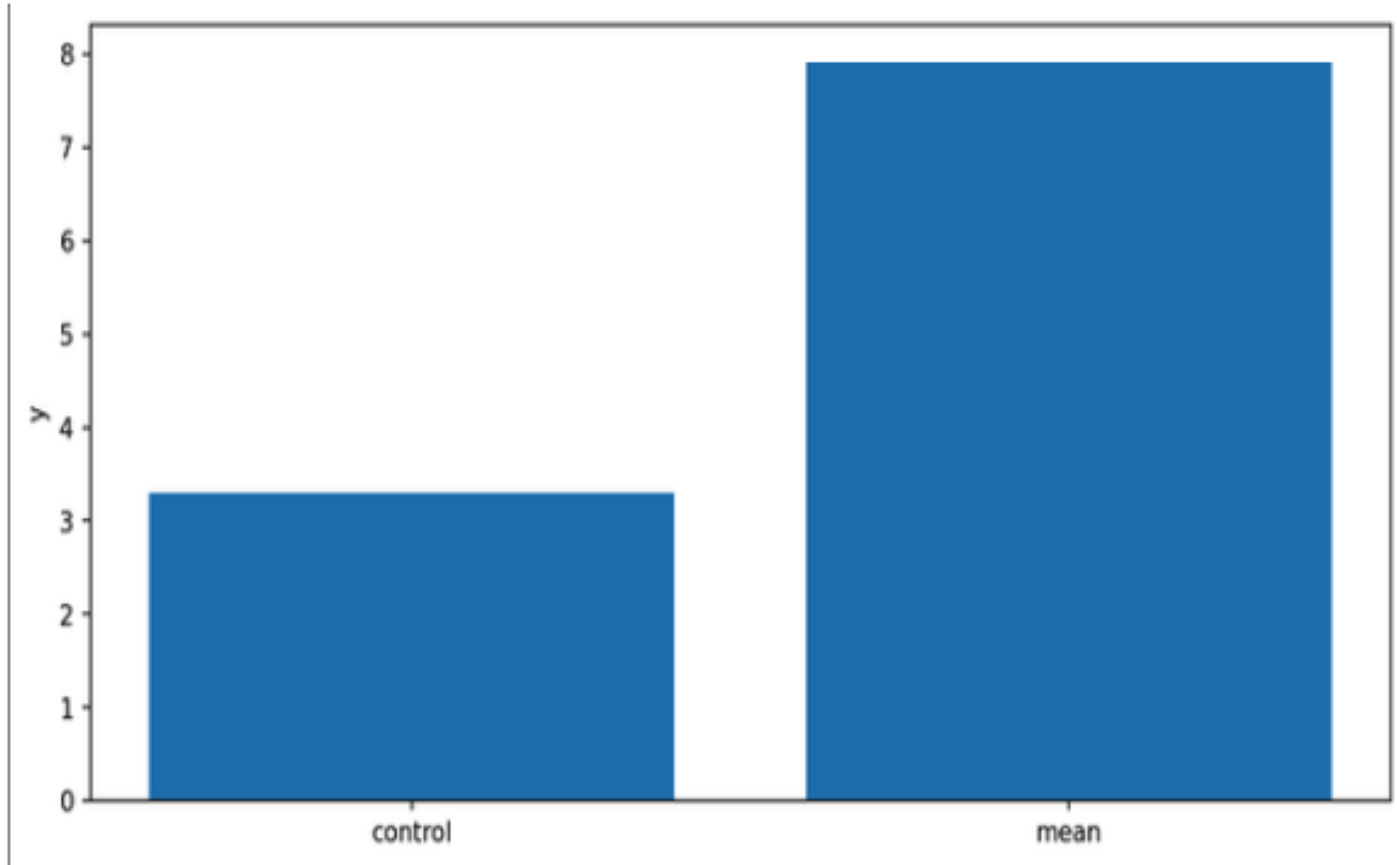


Breaking up a variable into categories

- what if I split y into control and test
- then the sample mean in each category could be compared

```
1 plt.figure()
2 plt.bar([1,2], [np.mean(y[category=='control']), np.mean(y[category=='test'])])
3 plt.ylabel('y')
4 plt.xticks([1,2], ['control', 'mean'])
5 plt.show()
```

Breaking up a variable into categories



Are these means actually different?

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 import scipy
4
5 category = np.array(['control', 'control', 'control', 'control', 'control',
6 y = np.array([1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 6, 7.1, 9.9, 10])
7
8 score, probability = scipy.stats.ttest_ind(y[category=='control'], y[cate
9
10 print(score)
11 print(probability)
```

Analyses for two continuous variables

Scatterplots

A **scatterplot** is a way to visualize the **relationship** between two variables.

- the x axis is for first variable
- the y axis is for second variable
- each dot is one observation

Scatterplot Example

- Once every two weeks for a year, we have recorded the number of air conditions sold and the number of ice cream cones sold on that day.
- How many observations would that be?
- What do you think the relationship would look like?

Scatterplot Example

- what is on x?
- what is on y?
- if you see a clear upward or downward slope, that means that the variables are related

n.b. a relationship does not imply causation!

Scatterplot Example

Find a positive relationship, negative relationship, no relationship, a non-linear relationship

Correlation

- Pearson Correlation, is defined:

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \times \sum_{i=1}^N (y_i - \bar{y})^2}}$$

- correlation of 1 is perfect positive slope
- correlation of -1 is perfect negative slope
- correlation of 0 is 0 or no slope

Correlation Example

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 x = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])
5 y = np.array([1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 6, 7.1, 9.9, 10])
6
7 plt.figure()
8 plt.scatter(x,y)
9 plt.xlabel('x')
10 plt.ylabel('y')
11 plt.show()
```

Correlation Example

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 x = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])
5 y = np.array([1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 6, 7.1, 9.9, 10])
6
7 numerator = np.sum((x - np.mean(x)) * (y - np.mean(y)))
8 denominator = (sum( (x - np.mean(x))**2 ) * sum( (y - np.mean(y))**2 )) *
9
10 r = numerator / denominator
11 print(r)
```

0.979131432462635

Correlation Example in practice

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 import scipy
4
5 x = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])
6 y = np.array([1.1, 2.2, 3.3, 4.4, 5.5, 6.6, 6, 7.1, 9.9, 10])
7
8 score, probability = scipy.stats.pearsonr(x, y)
9
10 print(score)
11 print(probability)
```